Cloud-Edge Selective Background Energy Constrained Filter for Real-Time Hyperspectral Target Detection

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***Abstract*— Constrained by the performance of edge devices and real time (RT) processing technology, the existing hyperspectral target detection algorithms often struggle to rapidly distinguish targets from complex background pixels during real-time detection. To address this issue, this article proposes a new real-time cloud-edge selective background energy constrained (CE-SBEC) hyperspectral target detection algorithm. This algorithm aims to obtain detection results in real-time after capturing new data. Moreover, it conducts in-depth analysis based on existing detection results and updates the algorithm's internal data to enhance its capabilities in terms of global background annihilation (GBA) and complex background suppression (CBS). Consequently, it improves the accuracy of subsequent real-time detection results. To enhance the resource utilization, this article deploys various task nodes of the algorithm separately on both the cloud and the edge, enabling collaborative execution of the CE-SBEC algorithm. In our context, edge devices are airborne equipment designed for the rapid acquisition and processing of data at the site of data collection, while cloud computing devices refer to high-performance computing clusters situated at a significant distance from the data collection site. Experimental results demonstrate that compared with existing detection algorithms, our newly proposed method achieves more accurate detection results while ensuring real-time performance.**

***Index Terms*— Cloud-edge collaboration, hyperspectral, real time (RT) detection, target detection.**

I. Introduction

HYPERSPECTRAL remote sensing is an important technique for Earth observation in the field of remote sensing [1]. Compared with traditional remote sensing images, hyperspectral images (HSIs) [2], [3], [4] offer higher spectral resolution and consist of tens to hundreds of narrow spectral bands. In addition, HSIs can be used to effectively extract the spectral characteristics of ground objects. Owing to these advantages, the hyperspectral remote sensing technology has been widely applied in various fields, such as agriculture, military, and medicine.

As one of the most important research directions in HSI processing [5], [6], [7], target detection [8], [9], [10], [11] can effectively utilize the spectral characteristics of different objects to explore the differences between the background and the target. Compared with traditional target detection techniques, the hyperspectral target detection can differentiate camouflaged targets and detect concealed objects.

In recent years, various hyperspectral target detection methods have been proposed. The most direct approach is to measure the spectral distance between the target and the pixels to be detected. The spectral angle mapper (SAM) [12] directly measures the angle between the spectrum of each pixel and the prior spectral signature of the target. The spectral information divergence (SID) algorithm [13] utilizes the relative entropy to explain the spectral information. However, due to the limitations of hyperspectral imaging technology in terms of spatial resolution, many targets appear as subpixel entities, which can hinder the performance of the above-mentioned algorithms in target detection tasks. Statistics-based methods, such as adaptive coherence estimator (ACE) [14] and constrained energy minimization (CEM) [15], [16] can effectively perform subpixel detection by utilizing background information during detection. However, these methods typically associate the statistical distribution of the background in an image with a Gaussian distribution. When the background becomes excessively complex, it can result in inaccurate estimation of the overall background information. Based on the sparse characteristics of HSIs, Chen et al. [17] proposed the sparse representation for target detection (STD) algorithm. This method assumes that each pixel in an HSI can be represented as a linear combination of a small number of spectral feature vectors from target and background pixels. By formulating this representation problem as a sparse-constrained optimization problem and solving it, an optimal sparse representation can be obtained. By comparing the reconstructed residuals, effective target detection in HSIs can be achieved.

To deeply explore the potential information within HSIs, certain researchers have employed deep learning techniques for HSI processing. Li et al. [18] present a new baseline network, referred to as low rank representation

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network (LRR-Net). This network synergizes the low rank representation (LRR) model with deep learning techniques. In [19], a two-stream convolutional network-based target detector was proposed for hyperspectral target detection. Zhang et al. [20] proposed a novel target detection framework for deep learning, called hyperspectral target detection network (HTD-Net). This framework overcomes the limited availability of training samples that generally exists in remote sensing problems. To address the problem that the complexity of actual scenes often limits the performance of traditional statistical methods which rely on model assumptions and thus the traditional machine learning methods rely on mapping functions with limited complexity, Rao et al. [21] proposed a Siamese transformer network for HSI target detection. However, deep learning methods may not be applicable for the specific problem studied in this work due to the following reasons. Before performing target detection in a new area, these deep learning algorithms need to acquire relevant data from that specific area for model training, since models trained on data from different areas cannot be directly used for target detection in the new area. This limitation can induce a significant delay from data acquisition to obtaining detection results, leading to the difficulty in fulfilling the real time (RT) requirement.

The hyperspectral target detection can identify objects which are difficult to be distinguished in traditional single band and multispectral images. However, certain challenges also arise. First, with the advancement of hyperspectral imaging technology, the resolution of HSIs continues to improve [22], [23], [24], [25], resulting in an exponential increase in data processing volumes. Second, in order to improve the detection accuracy, detection algorithms need to fully exploit the spectral information. Consequently, traditional high-precision detection algorithms often involve complex computational steps. These factors make it challenging for traditional hyperspectral detection algorithms to accurately detect targets in real time during practical applications.

To meet the demands of real-time detection, researchers need to design algorithms that can be integrated with hyperspectral data acquisition methods to achieve fast detection while capturing new data. Currently, push-broom imaging spectrometers [26], which utilize linear array detectors and acquire stereoscopic data lined-by-line (as shown in Fig. 1), are widely employed due to their high imaging quality. This article explores real-time detection algorithms for push-broom data acquisition scenarios.

In order to expedite the acquisition of results, the majority of researchers are committed to enhancing the speed of detection algorithms by reducing algorithm complexity or leveraging hardware acceleration. Macias et al. [27] performed hardware optimization of the automatic target detection and classification algorithm (ATDCA) using the Gram–Schmidt (GS) method on field-programmable gate arrays (FPGAs). This led to notable improvements in computational speed and power efficiency compared with the prior algorithm implementations. Du and Ren [28] introduced a rapid CLDA detection and classification algorithm for target detection. To address the issue of iteratively calculating the inversion of growing matrices and increasing matrix multiplications in the automatic target

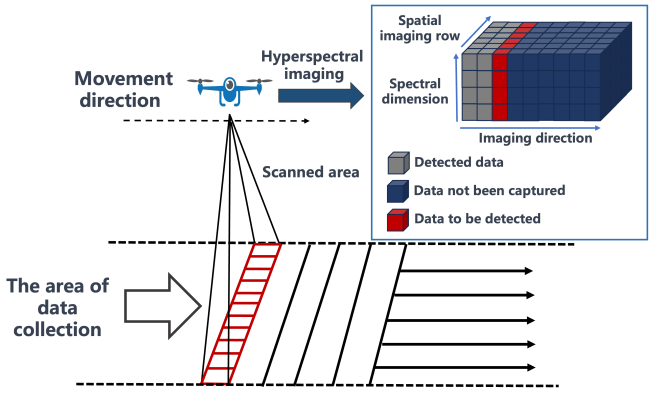

Fig. 1. Push-broom imaging spectrometer.

generation process based on an orthogonal subspace projector (ATGP-OSP) algorithm, Lei et al. [29] proposed the fast-ATGP algorithm, significantly improving the overall detection speed while maintaining target detection accuracy. They not only streamlined algorithm complexity but also implemented the algorithm on FPGAs for algorithmic acceleration. However, in the scenario of capturing data while simultaneously performing detection, the overall data information changes with the addition of new data. When the above-mentioned detection algorithms are applied to the newly captured data, a complete dataset recalibration is necessary. This process results in redundant data computations and precludes the attainment of real-time results.

To eliminate redundant data computations, Chang et al. [30] introduced an efficient real-time CEM algorithm. When new data are captured at the edge, there is no need to reprocess the entire dataset. Only the recently added data are analyzed. This strategy substantially minimizes the volume of data used in computations, guaranteeing real-time performance. However, in the process of performing detection while capturing data at the edge, algorithms can only utilize the data samples that have been accessed or captured for computation. They are unable to incorporate data samples that will be accessed or captured in the future. This limitation makes it challenging for algorithms to accurately calculate the overall background information during the detection process. On the other hand, real-time algorithms are typically deployed on edge devices with limited computing power to enable fast data retrieval and computation. Due to the constraints of edge device performance, the edge can only allocate a limited amount of computational resources. Therefore, real-time detection algorithms need to sacrifice a certain level of detection accuracy while improving the algorithm speed.

In general, the edge computing paradigm [31], [32], [33] can significantly reduce the latency and energy consumption required for data transmission over the network, whereas the cloud computing paradigm [34], [35], [36] allows for the detailed data analysis on the existing data. By combining the characteristics of edge computing and cloud computing paradigms [35], [37], [38], it becomes possible to execute computational tasks that provide high-precision real-time detection results. Today, cloud-edge computing has been widely used in various fields. In the most common cloud-edge computing

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paradigm, the edge offloads a part of computational tasks to the cloud, with both entities collaborating to accomplish a computational task [37]. However, this paradigm is only suitable for large-scale tasks involving simple computations. When a computational task or data cannot be divided, partial task offloading cannot be performed. In the literature [39], [40], complex algorithms are deployed at the cloud, while simple algorithms are deployed at the edge. Data are uploaded to the cloud for further computation only when the edge algorithm cannot fulfill the requirements. However, in this cloud-edge computing paradigm, the computational results from the cloud do not offer further feedback and optimization for the edge algorithms. To address this issue, Wang et al. [41] introduced a new cloud-edge computing paradigm. During the data acquisition process, the edge performs real-time computations. It filters and offloads data to the cloud for further computation. Then, the cloud sends back the (high accuracy) results to the edge for information updating. After receiving the results from the cloud, the edge updates the edge's detection model to enhance the accuracy of real-time computations.

To perform the real-time target detection tasks quickly and accurately, in this article, we propose a new selective background energy constrained (SBEC) algorithm aimed at enhancing the accuracy of real-time detection. During the process of simultaneously capturing data and conducting detection with a push-broom spectrometer, our algorithm can analyze the data and update the operators based on the previous detection results, enhancing the algorithm's capabilities of global background annihilation (GBA) and complex background suppression (CBS). To improve the resource utilization and ensure the algorithm's real-time performance, we further propose a cloud-edge SBEC (CE-SBEC) algorithm based on the SBEC algorithm. Here, we define edge devices as airborne equipment designed for the rapid acquisition and processing of data at the site of data collection. In contrast, cloud computing devices refer to high-performance computing clusters situated at a significant distance from the data collection site. The CE-SBEC algorithm cleverly utilizes the characteristics of both the cloud and the edge. During the execution of the proposed algorithm, the edge device conducts the real-time target detection on the captured data and periodically partitions the data based on the detection results. Subsequently, a portion of the data is uploaded to the cloud. The cloud leverages significant computational resources to further analyze the uploaded data and sends back the results to the edge device for updating purposes. This updating process aims to enhance the accuracy of the edge device in detecting subsequent captured data.

The main innovative contributions of the proposed approach can be summarized as follows.

1) To improve the detection accuracy, we propose an SBEC algorithm for target detection in data captured concurrently in a line-by-line fashion. This algorithm updates its parameters in real time by analyzing the existing detection results. On one hand, it can extract complex pure background signatures (endmembers) that the current filter cannot suppress, thereby enhancing the algorithm's GBA capability. On the other hand, this algorithm can continuously update background data and improve the CBS ability.
2) To improve the efficiency of the algorithm, we introduce a real-time CE-SBEC based on a cloud-edge collaborative framework. The algorithm maximizes the utilization of computational resources at both the cloud and the edge to ensure the overall real-time performance of the algorithm.
3) To avoid redundant data computations and streamline the computational flow in cloud-edge collaboration, we introduce a fast endmember extraction algorithm and a rapid edge update algorithm. While maintaining the original computational accuracy, these algorithms, respectively, boost the speed of endmember extraction in the cloud and the speed of data updates in the edge algorithm.

The rest of this article is organized as follows. Section II reviews the related works on hyperspectral real-time detection technology. The SBEC algorithm is described in Section III. Section IV provides a detailed description of the execution process of the CE-SBEC algorithm. We provide experimental results and analyses in Section V and conclude this article in Section VI.

II. Related Works

A. CEM Algorithm

Let us assume that there are N pixels in an HSI with L spectral bands. The ith pixel ($1 \leq i \leq N$) in the HSI is denoted by $r_i = [r_{i1}, r_{i2}, \ldots, r_{iL}]^{\mathrm{T}}$. The primary idea of the CEM algorithm is to design a finite impulse response (FIR) filter $w = [w_1, w_2, \ldots, w_L]^{\mathrm{T}}$ that allows the desired target signal $D = [D_1, D_2, \ldots, D_L]^{\mathrm{T}}$ to pass through. Therefore, the filter w must satisfy the following constraint:

$$D^{\mathrm{T}} w = \sum_{l=1}^{L} D_l w_l = 1. \tag{1}$$

Assuming that the filter w mentioned above corresponds to the input r_i, and the output is represented as y_i, we have

$$y_i = \sum_{l=1}^{L} w_l r_{il} = w^{\mathrm{T}} r_i. \tag{2}$$

The average energy of the filter's output for all pixels $\{r_1, r_2, \ldots, r_L\}$ is given by

$$\begin{aligned} E &= \frac{1}{N}\sum_{i=1}^{N} y_i^2 = \frac{1}{N}\sum_{i=1}^{N} \left(r_i^{\mathrm{T}} w\right)^{\mathrm{T}} r_i^{\mathrm{T}} w \\ &= \frac{1}{N}\sum_{i=1}^{N} w^{\mathrm{T}} r_i r_i^{\mathrm{T}} w = w^{\mathrm{T}} \left(\frac{1}{N}\sum_{i=1}^{N} r_i r_i^{\mathrm{T}}\right) w = w^{\mathrm{T}} R w \end{aligned} \tag{3}$$

where $R = (1/N)\sum_{i=1}^{N} r_i r_i^{\mathrm{T}}$ is the autocorrelation matrix. We aim to suppress the output energy of other pixels after passing through the filter. Therefore, the CEM algorithm is represented as the following linear constrained optimization

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problem:

$$\begin{cases} \min\left(w^{\mathrm{T}} R w\right) \\ D^{\mathrm{T}} w = 1. \end{cases} \tag{4}$$

Solving this extremum problem allows us to obtain the filter

$$w = \frac{R^{-1} D}{D^{\mathrm{T}} R^{-1} D}. \tag{5}$$

B. Real-Time CEM Algorithm

The computational load of the CEM algorithm primarily lies in the computation of the autocorrelation inverse matrix of the data. In real-time detection scenarios, whenever a new row of the hyperspectral data is captured, the autocorrelation inverse matrix needs to be updated. Therefore, RT-CEM simplifies the data computational approach. We represent the hyperspectral data of the lth row as ℓ_l and all data for the first m rows as $\ell(m)$. If each row of data has n pixels, then it can be expressed as follows:

$$R(\ell_l) = (1/n) \sum_{i=1}^{n} r_i^l \left(r_i^l\right)^{\mathrm{T}} \tag{6}$$

and

$$R(\ell(m)) = (1/m) \sum_{l=1}^{m} R(\ell_l). \tag{7}$$

Woodbury's identity states that the inverse of certain matrices with kth-order correction factors can be expressed using the inverse matrix of the original matrix and its kth-order correction factors. This principle is known as Woodbury's Lemma [42], [43], [44] and can be expressed as follows:

$$\left(A + UCV^{\mathrm{T}}\right)^{-1} = A^{-1} - A^{-1}U\left(C^{-1} + V^{\mathrm{T}}A^{-1}U\right)^{-1}V^{\mathrm{T}}A^{-1} \tag{8}$$

where A is an $s \times s$ matrix, U is an $s \times k$ matrix, C is a $k \times k$ matrix, and V^{T} is a $k \times s$ matrix. When C and V^{T} are the identity matrix, (8) can be simplified as follows:

$$(A + U)^{-1} = A^{-1} - A^{-1}U\left(A^{-1}U + I\right)^{-1}A^{-1}. \tag{9}$$

Clearly, if we know the value of A^{-1}, it would be much easier to calculate it using (8) than to find the inverse of the matrix $A + U$ directly. Let $A = ((m-1)/m)R((m-1))$, $U = (1/m)R(\ell_m)$ and we have

$$A^{-1}U = \frac{R(\ell(m-1))^{-1}}{m-1} R(\ell_m) = R_{m|m-1}. \tag{10}$$

According to (10), we can obtain

$$\begin{aligned} R(\ell(m))^{-1} &= \left(\frac{(m-1)}{m} R(\ell(m-1)) + \frac{R(\ell_m)}{m}\right)^{-1} \\ &= \frac{m}{m-1}\left\{R(\ell(m-1))^{-1} - R_{m|m-1}\left(R_{m|m-1} + I\right)\right\}^{-1} \\ &\quad \times R(\ell(m-1))^{-1}. \end{aligned} \tag{11}$$

When performing detection on the captured data for the mth row, the filter is

$$w_m^{\text{RT-CEM}} = \frac{d^{\mathrm{T}} R(\ell(m))^{-1}}{d^{\mathrm{T}} R(\ell(m))^{-1} d}. \tag{12}$$

According to (12), we get

$$\begin{aligned} w_m^{\text{RT-CEM}} &= \frac{m}{m-1}\left\{\left(\frac{d^{\mathrm{T}} R\left(\ell(m-1)^{-1} d\right)}{d^{\mathrm{T}} R(\ell(m))^{-1} d}\right) w_{m-1}^{\text{RT-CEM}} \right. \\ &\quad - \left(d^{\mathrm{T}} R(\ell(m))^{-1} d\right)^{-1} \\ &\quad \left. \times \left[d^{\mathrm{T}} R_{m|m-1}\left(R_{m|m-1} + I\right)^{-1} R\left(\ell(m-1)^{-1}\right)\right]\right\}. \end{aligned} \tag{13}$$

III. SBEC Algorithm

In real hyperspectral target detection scenarios, the data available are relatively limited at the initial detection stage. This limitation affects the ability of the filters in the RT-CEM algorithm to properly calculate the background information. In other words, some complex background pixels cannot be accurately suppressed by the filters. Consequently, this can lead to their misidentification as targets during the detection process. To mitigate the impact of complex background pixels on the filter and enhance real-time detection accuracy, in this section, we introduce a new SBEC algorithm.

A. Data Analysis in the SBEC Algorithm

The idea of the SBEC algorithm is originated from the background-annihilated target-constrained interference-minimized filter (BA-TCIMF) [45]. The BA-TCIMF algorithm consists of two stages. In the first stage, easily distinguishable background pixels are identified through model decomposition, and their autocorrelation inverse matrices are computed to perform global background elimination (GBA). In the second stage, challenging-to-detect background pixels are further suppressed based on the results of model decomposition, achieving CBS. However, the BA-TCIMF algorithm is more suitable for execution after capturing all hyperspectral data, rather than for real-time detection during data collection. In addition, in the initial stages (simultaneous data capture and detection), the limited amount of data leads to a decrease in the performance of GBA and CBS, affecting the overall detection results.

During real-time detection, the edge device is capable of continuously acquiring detection results. Different from BA-TCIMF algorithm, our SBEC algorithm can update the filters in the algorithm based on the previous detection results to enhance the execution of GBA and CBS. The process of analyzing detection result data consists of three steps.

1) The first step is data categorization. Based on the obtained detection results, we categorize all detected pixels into regular background pixels and complex pixels. Regular background pixels exhibit lower energy values after passing through the current filter, in which the filter can effectively suppress. On the other hand, complex pixels have higher energy values after passing through the current filter, and the filter cannot effectively suppress their energy.
2) The second step is the reconstruction of the autocorrelation matrix. We recalculate the autocorrelation inverse

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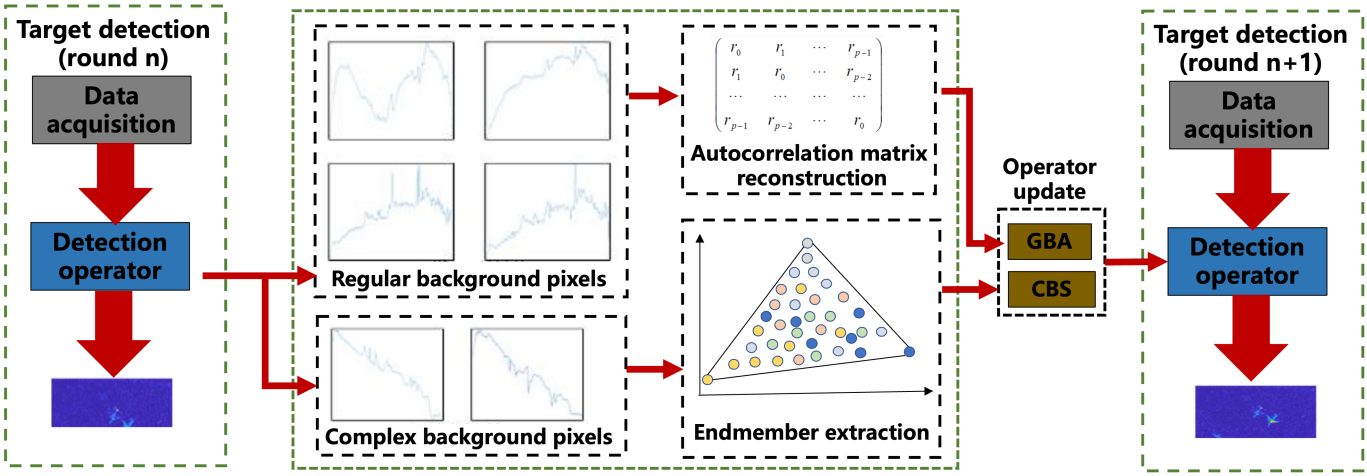

Fig. 2. Steps of the SBEC algorithm.

matrix based on the partitioned information of regular background pixels. Due to the absence of interference from complex background pixels and target pixels, utilizing the recalculated autocorrelation inverse matrix in the filter can more effectively suppress background pixels, enhancing the GBA capability of the algorithm.

3) The third step is the selection of complex background endmembers. Complex pixels include both target pixels and complex background pixels. Therefore, we start by extracting endmembers from all complex pixels to obtain the endmember information. Subsequently, we calculate the SAM [12], [46] values between each endmember and the target pixel. We utilize these values to scan the complex background endmembers and incorporate them into the filter for suppression. Typically, the number of complex background endmembers is much smaller than the number of complex background pixels. This approach helps enhancing the algorithm's capability in terms of CBS while ensuring computational efficiency.

B. Specific Steps of the SBEC Algorithm

The execution flow of the SBEC algorithm is illustrated in Fig. 2, and the specific steps are detailed in Algorithm 1. During the acquisition of the mth row hyperspectral data by the push-broom spectrometer, the filter is represented as follows:

$$w_m = R(m)^{-1}\left[D, U_{m-1}\right] \cdot \left(\left[D, U_{m-1}\right]^{\mathrm{T}} R(m)^{-1}\left[D, U_{m-1}\right]\right)^{-1} \begin{bmatrix} 1_{n_D \times 1} \\ 0_{n_U \times 1} \end{bmatrix}. \quad (14)$$

We assume that this result-based analysis update is performed after M rows of data have been captured. It is worth noting that, when $m \in [1, M]$, the algorithm lacks prior background knowledge to suppress. Hence, U_{m-1} is an empty set. When m is the multiple of M, it is necessary to perform pixel selection and data update.

We use $L_{(m-M+1)\sim m}$ to represent the pixels from $(m - M + 1)$th row to mth row in an HSI. The detection result of $L_{(m-M+1)\sim m}$ is represented as $B^{\text{result}}_{(m-M+1)\sim m}$. Based on the detected result $B^{\text{result}}_{(m-M+1)\sim m}$, the algorithm categorizes the pixels into complex pixels $X^{\text{complex}}_{(m-M+1)\sim m}$ and regular background pixels $X^{\text{background}}_{(m-M+1)\sim m}$. The regular background pixels $X^{\text{background}}_{(m-M+1)\sim m}$ can be effectively suppressed by the current filter w_m. The complex pixels $X^{\text{complex}}_{(m-M+1)\sim m}$ include the target pixel D and complex background pixels $X^{\text{complexbackground}}_{(m-M+1)\sim m}$. They cannot be effectively suppressed by the current filter w_m.

To extract information from complex background pixels and selectively suppress them in the subsequent real-time target detection, this article employs the virtual dimensionality (VD)

Algorithm 1 SBEC Algorithm

Input: The mth row of data $Data^e_m$ captured at the edge, background information U_m, target information D;

Output: Detection results B^{result}_m of data $Data^e_m$, matrix $R(m)^{-1}$, endmember U_m;

if $Data^e_m = \phi$ **then**
  return ϕ;
Calculate $R(m)^{-1}$; Use the filter w_m to perform detection on pixels $\{x_i | x_i \in Data^e_m\}$ and obtain the detection result B^{result}_m;
if $m\%M = 0$ **then**
  Filter out the regular background pixels $X^{background}_{(m-M+1)\sim m}$ and complex pixels $X^{complex}_{(m-M+1)\sim m}$ based on the detection results from the $m - Mth$ to mth rows;
  Calculate the autocorrelation inverse matrix $R(m)^{-1}_{background}$ of $X^{background}_{1\sim m}$ and update it as $R(m)^{-1}$;
  Combine the complex pixels $X^{complex}_{(m-M+1)\sim m}$ selected this time with all the previously selected complex pixels to obtain $X^{complexbackground}_{1\sim m}$. Perform endmember extraction on $X^{complexbackground}_{1\sim m}$, filter the complex background endmembers $U^{complexbackground}_{1\sim m}$ by calculating SAM values. Let $U_m = U^{complexbackground}_{1\sim m}$.
return B^{result}_m, $R(m)^{-1}$, U_m;

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algorithm [47] to estimate the number of endmembers and the N-FINDR algorithm [48], [49], [50] to extract endmembers from all filtered complex background pixels $X_{1\sim m}^{\text{complexbackground}}$. After the endmember extraction, SAM values are calculated between the extracted endmembers and the target spectral information D. Endmembers with SAM values less than 0.98 are recognized as complex background endmembers $U_{1\sim m}^{\text{complexbackground}}$ and integrated into the filter for updating.

To avoid interference from target pixels in the overall background suppression, we update the matrix $R(m)^{-1}$. We replace the autocorrelation inverse matrix of the entire captured dataset with the autocorrelation inverse matrix of all known background pixels $X_{1\sim m}^{\text{background}}$, denoted by $R(m)_{\text{background}}^{-1}$. This helps preventing interference from complex pixels on the overall filter accuracy, enhancing the algorithm's suppression effect on regular pixels.

C. Validity Proof for the Data Update

We denote the filter as w. The data to be detected are denoted by $L_l = \{r_1^l, r_2^1, \ldots, r_n^l\}$, this is the lth row of the HSI. Pixel $r_i^l (1 \le i \le n)$ can be represented as follows:

$$r_i^l = \alpha_i U + \beta_i B \tag{15}$$

where α_i and β_i represent the corresponding coefficients. U represents the complex background endmembers and B represents the regular background endmembers. If $\alpha_i U$ constitutes the main components, the pixel r_i^l is a complex background pixel. If $\beta_i B$ constitutes the main components, the pixel r_i^l is a regular background pixel.

When the pixel r_i^l passes through the filter w, we have

$$y = \left(r_i^l\right)^{\mathrm{T}} w = \alpha_i U^{\mathrm{T}} w + \beta_i B^{\mathrm{T}} w. \tag{16}$$

Due to the implementation of CBS, we have

$$U^{\mathrm{T}} w = 0. \tag{17}$$

Due to the execution of GBA, the value of $\beta_i B^{\mathrm{T}} w$ is also small. In summary, whether the pixel r_i^l belongs to the category of regular or complex background pixels, the filter w can more effectively suppress their energy, preventing that it is mistakenly detected as the target pixel.

IV. Real-Time Hyperspectral Target Detection Based on Cloud-Edge Collaboration

A. Cloud-Edge Collaboration in CE-SBEC Algorithm

In the traditional cloud computing process, when hyperspectral data are captured at the edge, they need to be transmitted to the cloud for centralized processing [51], [52]. In contrast, existing real-time algorithms are deployed at the edge. They perform real-time computation while capturing the data, eliminating the need to transmit the data to the cloud. With the rise of cloud-edge collaboration technology, leveraging both the cloud and the edge for target detection can significantly improve the efficiency of computing resources, ensuring the real-time performance of algorithms.

In comparison with the RT-CEM algorithm, our SBEC algorithm cam potentially make significant advancements in the task of improving detection accuracy. However, when processing high-dimensional hyperspectral data of the same size, the SBEC algorithm also incurs in more computational latency than the RT-CEM algorithm. Constrained by the limited computational resources at the edge, the SBEC algorithm faces challenges in ensuring real-time detection. To enhance the computational speed of the SBEC algorithm, we deploy it in the cloud-edge environment proposed in [41].

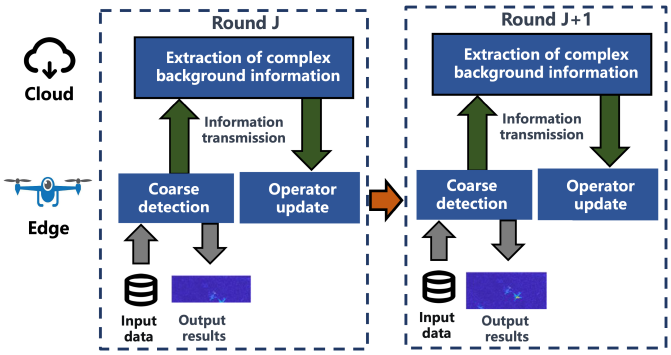

Fig. 3. Cloud-edge collaboration approach in the CE-SBEC algorithm.

We assume that the CE-SBEC algorithm consists of multiple rounds of cloud-edge collaboration. Fig. 3 illustrates the data computation and transmission between the cloud and the edge during the Jth and $(J+1)$th cloud-edge interactions. In the beginning of the $(J+1)$th round of cloud-edge collaboration, the edge uploads the complex pixels filtered in the Jth round to the cloud. Typically, during the process of target detection, the number of target pixels is relatively small compared with other background pixels. Therefore, each time we only transmit $1/k$ of the pixels to the cloud based on previous results, alleviating the transmission pressure on network bandwidth during cloud-edge interaction. In the $(J+1)$th round of collaboration, while the edge captures and detects in real-time, the cloud further analyzes the complex pixels uploaded by the edge in the previous cloud-edge interaction. After both the edge and the cloud complete their respective computations, they communicate again. At this point, the cloud sends the filtered complex background pixel information to the edge, and the edge performs data updates for the filter.

According to [41], the computational delay T_J of the cloud-edge collaborative model during the Jth round of cloud-edge collaboration can be expressed as follows:

$$T_J = \max\left(t_J^{\text{detect,edge}} + t_J^{\text{update,edge}}, t_J^{\text{comp,cloud}} + t_J^{\text{cor}}\right) \tag{18}$$

where $t_J^{\text{comp,cloud}}$ represents the time spent by cloud computing in the Jth round, $t_J^{\text{detect,edge}}$ represents the time spent by edge computing in the Jth round, t_j^{cor} represents the time for data transmission between cloud and edge in the Jth round, and $t_J^{\text{update,edge}}$ represents the time spent by edge computing to update the correlation matrix based on the segmentation results in the Jth round. Compared with real-time CEM algorithms we can see that, at the edge, in addition to the delay in real-time detection, there is only an added delay for the data update step.

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Algorithm 2 Fast Edge Update Algorithm

Input: Matrix $R(m)^{-1}$ and complex pixels $X^{complex}_{(m-M+1)\sim m}$;
Output: The updated matrix $R(m)^{-1}$;
1 Calculate the correlation matrix $r_{complex}$ for the complex pixels $X^{complex}_{(m-M+1)\sim m}$;
2 Equation (21) is used to obtain the updated matrix $R(m)^{-1}_{background}$ at the edge;
3 Let $R(m)^{-1} = R(m)^{-1}_{background}$;
4 **return** $R(m)^{-1}$;

B. Edge Updating Algorithm

After categorizing the background pixels and suspicious pixels, we need to recalculate the correlation inverse matrix $R(m)^{-1}_{\text{background}}$ for all known background pixels at the edge. Therefore, each update at the edge consumes a considerable amount of time. To reduce the time $t_J^{\text{update,edge}}$, we split the formula of $R(m)$ as follows:

$$\begin{aligned} R(m) &= \frac{1}{m}\sum_{i=1}^{m} x_i x_i^{\mathrm{T}} \\ &= \frac{1}{m}\left(\sum_{i\in I_a} x_i x_i^{\mathrm{T}} + \sum_{i\notin I_a} x_i x_i^{\mathrm{T}}\right) \\ &= \frac{m-k}{m} R(m)_{\text{background}} + \frac{k}{m} r_{\text{complex}} \end{aligned} \tag{19}$$

where I_a is used to represent the position indices of the suspicious pixels filtered based on the results $B^{\text{result}}_{(m-M+1)\sim m}$, and r_{complex} represents the correlation matrices of these suspicious pixels. According to (21), we can further derive

$$R(m)_{\text{background}} = \frac{m}{m-k} R(m) - \frac{k}{m-k} r_{\text{complex}}. \tag{20}$$

According to Woodbury's constant equation, we can obtain

$$\begin{aligned} R(m)^{-1}_{\text{background}} &= \left(\frac{m}{m-k} R(m) - \frac{k}{m-k} r_{\text{complex}}\right)^{-1} \\ &= \frac{m-k}{m}\left\{R(m)^{-1} + R^*\left(I - R^*\right)^{-1} R(m)^{-1}\right\} \end{aligned} \tag{21}$$

where $R^* = (k/(m-k))R(m)^{-1} r_{\text{complex}}$. According to (21), we have significantly reduced the complexity of the update step. Algorithm 2 describes the specific steps of the edge update algorithm.

C. Fast Endmember Extraction Algorithm at the Cloud

The N-FINDR algorithm belongs to the category of pure pixel geometric methods for linear spectral unmixing [53]. It identifies endmembers by inflating a simplex within the data, finding the set of pure pixels that define the simplex with the maximum possible volume. Endmembers are the vertices of the simplex. In the spectral dimension, the volume defined by the simplex made up of the purest pixels is greater than any other volume defined by combinations of (nonpure) pixels. As shown in Fig. 4, u_1–u_3 represent the endmembers.

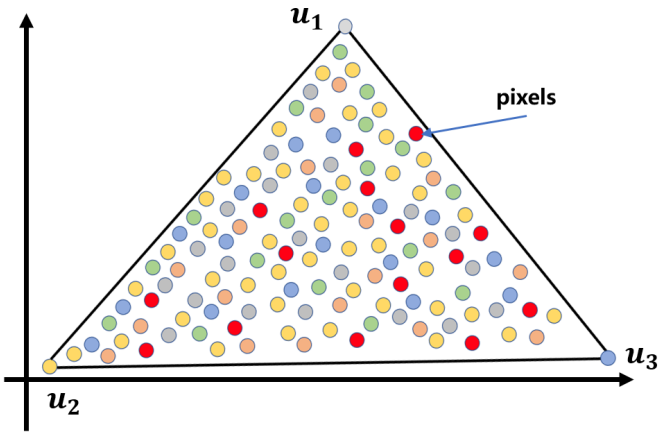

Fig. 4. Endmember extraction with all pixels.

Algorithm 3 Fast Endmember Extraction Algorithm

Input: The previous endmember extraction results $U^{complexbackground}_{1(m-M)}$ and data $X^{complex}_{(m-M+1)\sim m}$;
Output: complex background endmembers $U^{complexbackground}_{1\sim m}$;
1 **if** $X^{complex}_{(m-M+1)\sim m} = \phi$ **then**
2 **return** ϕ;
3 Merge $U^{complexbackground}_{1\sim(m-M)}$ and $X^{complex}_{(m-M+1)\sim m}$, let $X^{cloud}_{1\sim m} = [U^{complexbackground}_{1\sim(m-M)}, X^{complex}_{(m-M+1)\sim m}]$;
4 Use VD algorithm to estimate the number of endmembers for the data. Then use the N-FINDR algorithm to get $U^{cloud}_{1\sim m}$.
5 Calculate SAM between d and each endmember $U^{cloud}_{1\sim m}$. Endmembers with calculation results less than 0.98 are considered as complex background endmembers $U^{complexbackground}_{1\sim m}$.
6 **return** $U^{complexbackground}_{1\sim m}$;

They form a simplex with the maximum volume. In real-time detection, as more data are received at the cloud, performing endmember extraction each time new data are received results in a significant waste of resources and redundant computations. Even with the immense computational resources available in the cloud, quick retrieval of endmembers may be difficult. The computation time of $t_J^{\text{comp,cloud}} + t_J^{\text{cor}}$ is lengthy and affects the overall delay in cloud-edge collaboration.

To address this issue, we propose a fast implementation of the N-FINDR algorithm. This method takes the characteristics of geometric endmember extraction techniques into account. Fig. 5 illustrates the process of endmember extraction at the cloud when a total of six rounds of pixel transmission occur between the cloud and the edge device. The fundamental idea is that, whenever the cloud receives new complex endmembers, it merges these new complex endmembers with the complex background endmembers extracted in the previous round (rather than merging them with all previously uploaded complex endmembers). The endmember extraction is then performed on the merged data. This approach avoids redundant endmember extraction on hyperspectral pixels, significantly reducing the computation time at the cloud. Algorithm 3

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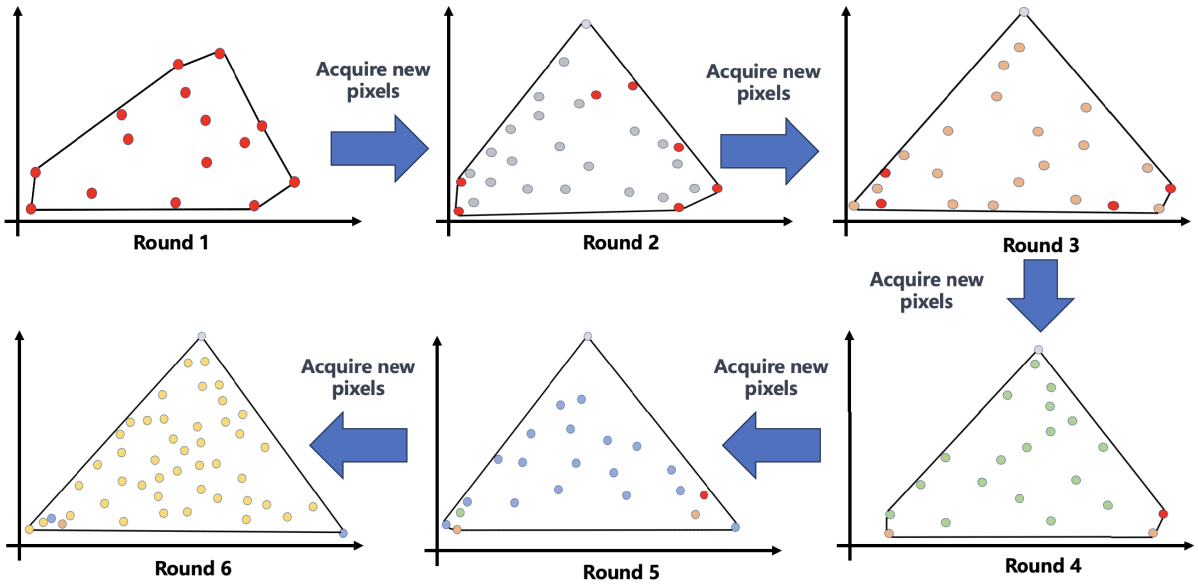

Fig. 5. Fast endmember extraction.

Algorithm 4 CE-SBEC Algorithm

Input: The mth row of data $Data_m^e$ captured at the edge, background information U_m, target information D;

Output: Detection results B_m^{result} of data $Data_m^e$, matrix $R(m)^{-1}$, endmember U_m;

if $Data_m^e = \phi$ **then**
  return ϕ;
Calculate $R(m)^{-1}$; Use the filter w_m to perform detection on pixels $\{x_i | x_i \in Data_m^e\}$ and obtain the detection result B_m^{result};
The edge calculates $R(m)^{-1}$ and uses the filter w_m to perform detection on pixels $\{x_i | x_i \in Data_m^e\}$ to obtain the detection result B_m^{result};
if $m\%M = 0$ **then**
  The edge filters out the regular background pixels $X_{(m-M+1)\sim m}^{background}$ and complex pixels $X_{(m-M+1)\sim m}^{complex}$ based on the detection results from $m - Mth$ to mth rows;
  The edge uploads the complex pixels $X_{(m-M+1)\sim m}^{complex}$ to the cloud;
  The cloud computes the matrix $r_{complex}$ based on $X_{(m-M+1)\sim m}^{complex}$;
  The cloud utilizes Algorithm 3 to compute $U_{1\sim m}^{complexbackground}$, let $U_m = U_{1\sim m}^{complexbackground}$;
  The cloud sends the data U_m and $r_{complex}$ to the edge;
  The edge executes Algorithm 2 to obtain the updated $R(m)^{-1}$.
return B_m^{result}, $R(m)^{-1}$, U_m;

describes the specific steps of the N-FINDR algorithm based on the causal computing approach. In summary, the detailed steps of the CE-SBEC algorithm are as given in Algorithm 4.

V. Experimental Results

A. Dataset Description

To test the effectiveness and feasibility of the proposed algorithm, we selected three real datasets for comparison purposes (see Fig. 6).

The detailed information are presented in Table I. Data 1 and Data 2 consist of HSIs from San Diego airport, collected using an Airborne Visible-Infrared Imaging Spectrometer (AVIRIS). The images comprise 224 spectral bands with a spatial resolution of 3.5 m/pixel. During the preprocessing stage, we removed water vapor bands and some contaminated bands (bands 1–6, 33–35, 94–97, 107–113, 153–166, and 221–224). A total of 186 usable bands were retained for the experiments. Data 1 represents a 100 × 100 region in the top left corner of the entire image, consisting of the background portion with an apron, rooftop, and a small amount of vegetation. The target points are three airplanes on the apron, totaling 57 pixels. Data 2 represents a 100 × 100 region in the middle of the entire image, containing 120 pixels. Samples for Data 3 were also manually extracted from maps downloaded from the AVIRIS website, and a noise level estimation method from [54] and [55] was employed to remove noise bands from the original images.

B. Performance Metrics

In this work, receiver operating characteristic (ROC) curves [56] and the area under the curve (AUC) values are used as evaluation criteria for experimental results. We utilize the AUC scores [denoted by $AUC_{(PD,PF)}$] based on false positive rate (PF) and detection rate (PD) to assess the performance of the detection algorithm. The closer the $AUC_{(PD,PF)}$ is to 1.0, the better the detection results. We use the AUC score (denoted by $AUC_{(PF,\tau)}$) based on PF rate and threshold τ to evaluate the background suppression performance. When $AUC_{(PF,\tau)}$ is 0, it means that no background pixels are incorrectly

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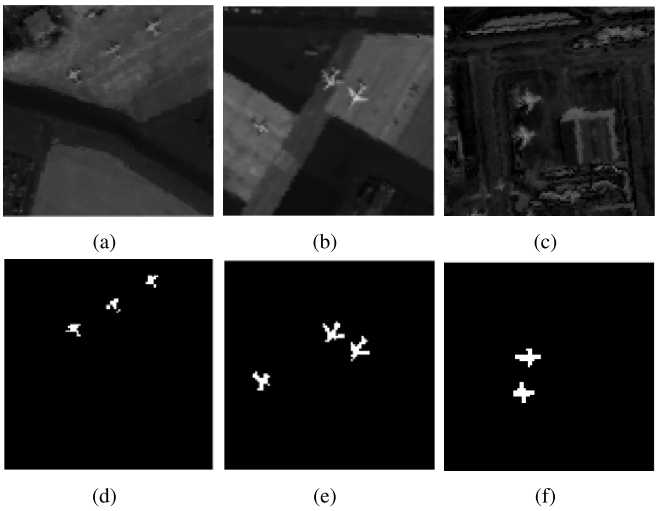

Fig. 6. Dataset. (a) Data 1 (original image). (b) Data 2 (original image). (c) Data 3 (original image). (d) Data 1 (ground-truth map). (e) Data 2 (ground-truth map). (f) Data 3 (ground-truth map).

TABLE I
DETAILED PARAMETERS OF THE CONSIDERED DATASETS

Dataset	Data acquisition location	Spatial resolution	Sensor
Data1	San Diego	3.5m	AVIRIS
Data2	San Diego	3.5m	AVIRIS
Data3	Los Angeles	7.1m	AVIRIS

detected as targets. The AUC score based on threshold τ and detection rate (PD) is used to evaluate target detectability. It is denoted by $\mathrm{AUC}_{(\mathrm{PD},\tau)}$. A higher $\mathrm{AUC}_{(\mathrm{PD},\tau)}$ value indicates better performance.

To evaluate the detector more comprehensively, using a joint target detection and background suppression measurement is more suitable than using PD and PF solely and separately. By taking the factor $\mathrm{AUC}_{(\mathrm{PD},\tau)}$ into $\mathrm{AUC}_{(\mathrm{PD},\mathrm{PF})}$ account, we use $\mathrm{AUC}_{\mathrm{TD}}$ to further assess the performance of the target detectability as follows:

$$0 \leq \mathrm{AUC}_{\mathrm{TD}} = \mathrm{AUC}_{(\mathrm{PD},\mathrm{PF})} + \mathrm{AUC}_{(\mathrm{PD},\tau)} \leq 2. \quad (22)$$

By taking the factor $\mathrm{AUC}_{(\mathrm{PF},\tau)}$ into $\mathrm{AUC}_{(\mathrm{PD},\mathrm{PF})}$ account, we use the $\mathrm{AUC}_{\mathrm{BS}}$ derived from both the 2-D ROC curve of (PD, PF) and the 2-D ROC curve of $\mathrm{AUC}_{(\mathrm{PF},\tau)}$ to further assess the performance of the background suppression as follows:

$$-1 \leq \mathrm{AUC}_{\mathrm{BS}} = \mathrm{AUC}_{(\mathrm{PD},\mathrm{PF})} - \mathrm{AUC}_{(\mathrm{PF},\tau)} \leq 1. \quad (23)$$

We also use $\mathrm{AUC}_{\mathrm{ODP}}$ to evaluate the overall detection performance of the algorithm

$$0 \leq \mathrm{AUC}_{\mathrm{ODP}} = \mathrm{AUC}_{(\mathrm{PD},\tau)} + \left(1 - \mathrm{AUC}_{(\mathrm{PF},\tau)}\right) \leq 2. \quad (24)$$

One key factor is that greater values of $\mathrm{AUC}_{(\mathrm{PD},\mathrm{PF})}$ and $\mathrm{AUC}_{(\mathrm{PD},\tau)}$ indicate better detection performance. Conversely, a reduced $\mathrm{AUC}_{(\mathrm{PF},\tau)}$ value signifies enhanced background suppression and, consequently, improved detection performance. As a result, (24) combines the AUC values derived from the three 2-D ROC curves to yield a single quantitative metric for assessing the overall detection performance.

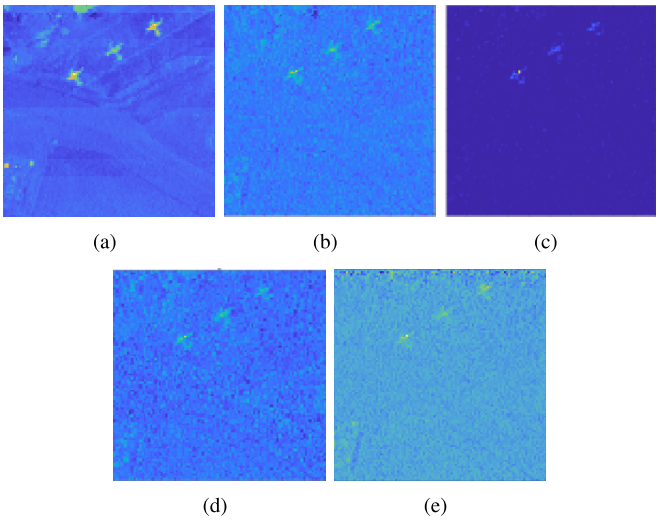

Fig. 7. Detection performance of different algorithms for Data 1. (a) SBEC algorithm. (b) RT-CEM algorithm. (c) ACE algorithm. (d) BA-TCIMF algorithm. (e) TD-TSE algorithm.

C. Detection Performance

In order to evaluate the detection accuracy, the proposed algorithm will be compared with the following target detection algorithms: RT-CEM algorithm, ACE algorithm [57], BA-TCIMF algorithm, and target detection method through tree-structured encoding (TD-TSE) [58]. To assess the real-time detection accuracy, we executed these algorithms in the context of real-time data capture and detection: only one line of data is detected at one time, and, during detection, the data involved in the calculations are limited to that line and the preceding pixels. When processing simultaneously capturing data and conducting detection with a push-broom spectrometer, conventional deep learning algorithms cannot access additional data from the same area in advance for model training. Furthermore, the data captured in real-time detection is dynamic and continuously increasing over time. The limited computational capacity of edge devices hinders the real-time update of the deep learning model based on newly captured data and conducting detection. For this reason, deep learning-based algorithms for hyperspectral target detection are not considered in the comparative experiments. The SBEC algorithm performs result analysis and algorithm updates every five lines of data detected. The simulation environment in the experiment is a 64-bit system, the CPU is Intel[1] Core[2] i5-7300HQ with 2.50 GHz, the system memory is 8G, and the simulation software is MATLAB R2018b.

For Data 1, Fig. 7 presents the experimental results of all the compared algorithms, demonstrating the excellent detection performance and observability of the SBEC algorithm proposed in this work. In the real-time detection scenario, distinguishing the target from the background poses a challenge for other algorithms.

Furthermore, Fig. 8 illustrates the corresponding ROC curves for experimental results. Table II provides a comprehensive view by listing the $\mathrm{AUC}_{(\mathrm{PD},\mathrm{PF})}$, $\mathrm{AUC}_{(\mathrm{PD},\tau)}$, $\mathrm{AUC}_{(\mathrm{PF},\tau)}$,

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TABLE II
AUC FOR THE DETECTION ALGORITHMS IN FIG. 7

Algoritms	$AUC_{(PD,PF)}$	$AUC_{(PD,\tau)}$	$AUC_{(PF,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{ODP}
CE-SBEC	**0.9976**	**0.6882**	0.1786	**1.6858**	0.8190	**1.5095**
RT-CEM	0.9527	0.3761	0.2186	1.3289	0.7341	1.1745
ACE	0.8973	0.1130	**0.0082**	1.0102	**0.8891**	1.1047
BA-TCIMF	0.9325	0.4068	0.2589	1.3393	0.6736	1.1479
TD-TSE	0.9452	0.5406	0.3906	1.4868	0.5546	1.1510

TABLE III
AUC FOR THE DETECTION ALGORITHMS IN FIG. 9

Algoritms	$AUC_{(PD,PF)}$	$AUC_{(PD,\tau)}$	$AUC_{(PF,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{ODP}
CE-SBEC	**0.9940**	0.4962	0.1160	**1.4850**	**0.8780**	**1.3797**
RT-CEM	0.9888	0.4740	0.1757	1.4680	0.8131	1.2983
ACE	0.7953	0.1060	**0.0132**	0.9014	0.7821	1.0928
BA-TCIMF	0.7602	**0.6327**	0.5327	1.3930	0.2275	1.1000
TD-TSE	0.8291	0.5068	0.4309	1.3360	0.3982	1.0758

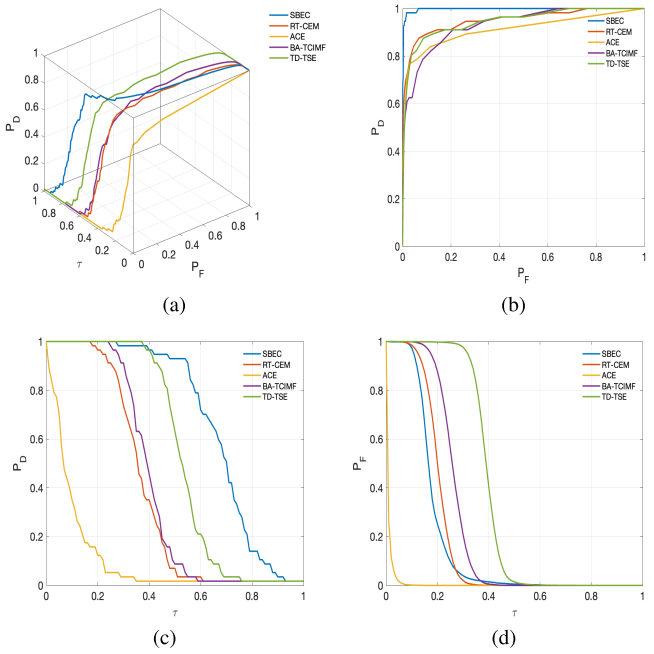

Fig. 8. ROC curves obtained by different methods for Data 1. (a) 3-D ROC curves. (b) ROC curves of (PD, PF). (c) ROC curves of (PD, τ). (d) ROC curves of (PF, τ).

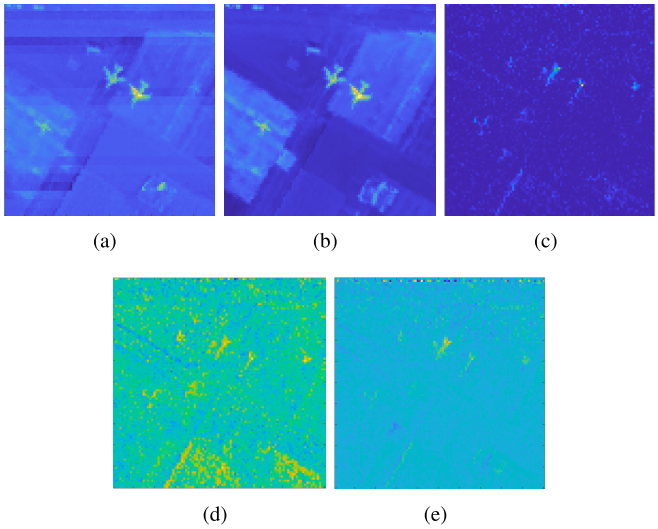

Fig. 9. Detection performance of different algorithms for Data 2. (a) SBEC algorithm. (b) RT-CEM algorithm. (c) ACE algorithm. (d) BA-TCIMF algorithm. (e) TD-TSE algorithm.

AUC_{TD}, AUC_{BS}, and AUC_{ODP} values of the compared algorithms.

It can be observed that the SBEC algorithm exhibits the highest $AUC_{(PD,PF)}$ value. In comparison with the RT-CEM algorithm, both $AUC_{(PD,\tau)}$ and $AUC_{(PF,\tau)}$ for the SBEC algorithm have increased. The ACE algorithm has the lowest $AUC_{(PF,\tau)}$ value and the highest AUC_{BS} value, indicating its superior background suppression performance compared to other algorithms. However, $AUC_{(PD,\tau)}$ and AUC_{TD} values for the ACE algorithm are quite low. Overall, the AUC_{ODP} value for the ACE algorithm is the lowest, at only 1.1047. In contrast, the SBEC algorithm achieves the highest AUC_{ODP} value, indicating its outstanding performance in detection.

Fig. 9 shows the experimental results of the comparative algorithms on Data 2, while Fig. 10 illustrates the corresponding ROC curves for each of these results. In contrast to Data 1, Data 2 showcases a relatively homogeneous distribution of background pixels, as a result, the detection results of the SBEC algorithm closely resemble those of the RT-CEM algorithm.

From Table III, it is evident that, when considering solely target detection capability and background suppression performance, the BA-TCIMF algorithm boasts the highest $AUC_{(PD,\tau)}$ value at 0.6327, whereas the ACE algorithm records the lowest $AUC_{(PF,\tau)}$ value at 0.0132. However, the relatively high $AUC_{(PF,\tau)}$ value for the BA-TCIMF algorithm implies

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TABLE IV
AUC FOR THE DETECTION ALGORITHMS IN FIG. 11

Algoritms	$AUC_{(PD,PF)}$	$AUC_{(PD,\tau)}$	$AUC_{(PF,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{ODP}
CE-SBEC	**0.9447**	0.6049	0.2165	**1.5496**	0.7282	**1.2553**
RT-CEM	0.9138	0.3590	0.3496	1.2728	0.5642	1.1425
ACE	0.8557	0.0923	**0.0050**	0.9480	**0.8507**	1.0873
BA-TCIMF	0.8812	0.4926	0.3205	1.3738	0.5607	1.1721
TD-TSE	0.8807	**0.6210**	0.4446	1.5017	0.4361	1.1764

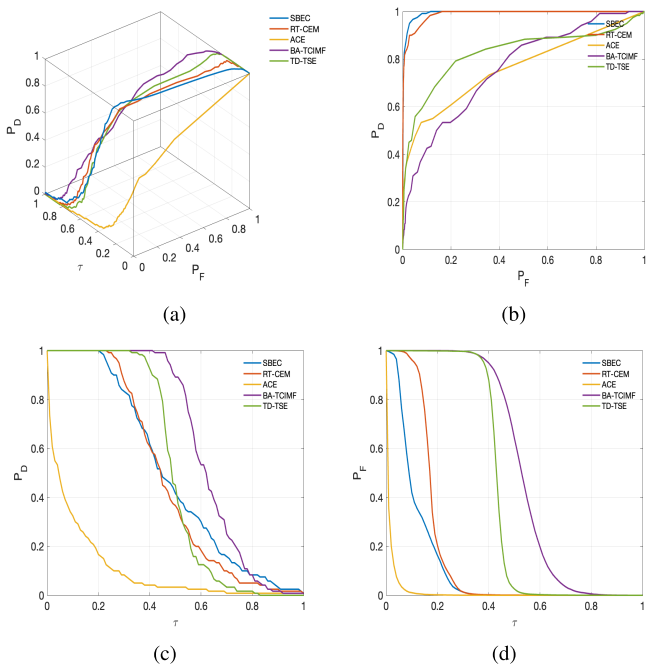

Fig. 10. ROC curves obtained by different methods for Data 2. (a) 3-D ROC curves. (b) ROC curves of (PD, PF). (c) ROC curves of (PD, τ). (d) ROC curves of (PF, τ).

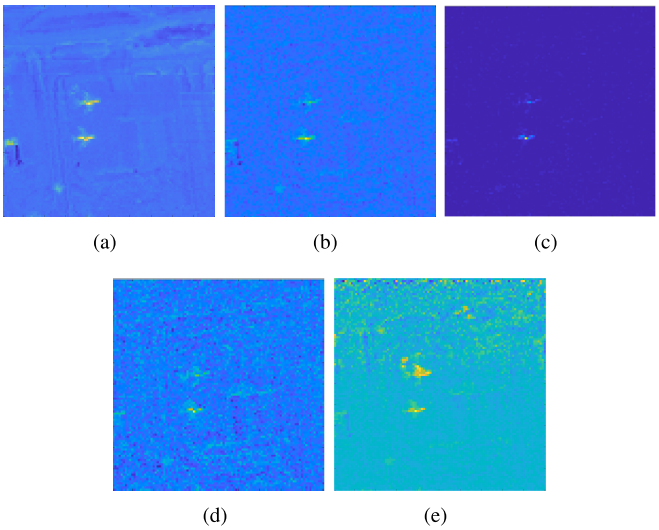

Fig. 11. Detection performance of different algorithms for Data 3. (a) SBEC algorithm. (b) RT-CEM algorithm. (c) ACE algorithm. (d) BA-TCIMF algorithm. (e) TD-TSE algorithm.

TABLE V
COMPUTATIONAL LATENCY OF SBEC AND CE-SBEC ALGORITHMS

Algorithms	Data1	Data2	Data3
SBEC	8s	7s	10s
CE-SBEC	0.3s	0.28s	0.34s

TABLE VI
$AUC_{(PD,PF)}$ VALUES FOR SBEC ALGORITHM AND CE-SBEC ALGORITHM

Algorithms	Data1	Data2	Data3
SBEC	0.9976	0.9940	0.9447
CE-SBEC	0.9910	0.9943	0.9421

its challenges in effectively suppressing the background. The ACE algorithm, on the other hand, exhibits a relatively low $AUC_{(PD,\tau)}$ value of 0.1060, as evident from Fig. 9, highlighting its suboptimal performance in target detection. The SBEC algorithm achieves the highest values in $AUC_{(PD,PF)}$, AUC_{TD}, AUC_{BS}, and AUC_{ODP}, underscoring its exceptional detection performance.

Fig. 11 displays the experimental results of the comparative algorithms on Data 3. Fig. 12 depicts the respective ROC curves for each of these results. It is evident that the detection algorithm proposed in this article continues to exhibit strong detection performance and observability. In comparison with the detection results of other algorithms, there is a small, relatively bright area on the left side of the image produced by the SBEC algorithm. This is because, during the previous stages of the detection process, this particular type of background pixel had not been encountered, making it challenging to preemptively suppress, which resulted in higher data values. From Table IV, it is evident that, compared with the RT-CEM algorithm, the SBEC algorithm achieves a higher $AUC_{(PD,\tau)}$ value and a lower $AUC_{(PF,\tau)}$ value. Furthermore, AUC_{ODP} value for the SBEC algorithm remains the highest among the comparative algorithms.

D. Real-Time Performance Analysis of the Proposed Algorithm

To evaluate the computing latency of our proposed algorithm, we conducted extensive experiments on a Jetson AGX Orin development board and a cloud server. The Jetson AGX Orin development board is equipped with a 12-core Cortex-A78 arm processor, a 2048-core GPU with 64 Tensor

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TABLE VII
COMPUTATIONAL LATENCY OF THE CE-SBEC ALGORITHM

Hyperspectral Data	t_J^{cor}	$t_J^{detect,edge}$	$t_J^{comp,cloud}$	$t_J^{update,edge}$	T_J
real-data1	62ms	270ms	190ms	30ms	300ms
real-data2	51ms	250ms	201ms	30ms	280ms
real-data3	74ms	300ms	241ms	40ms	340ms

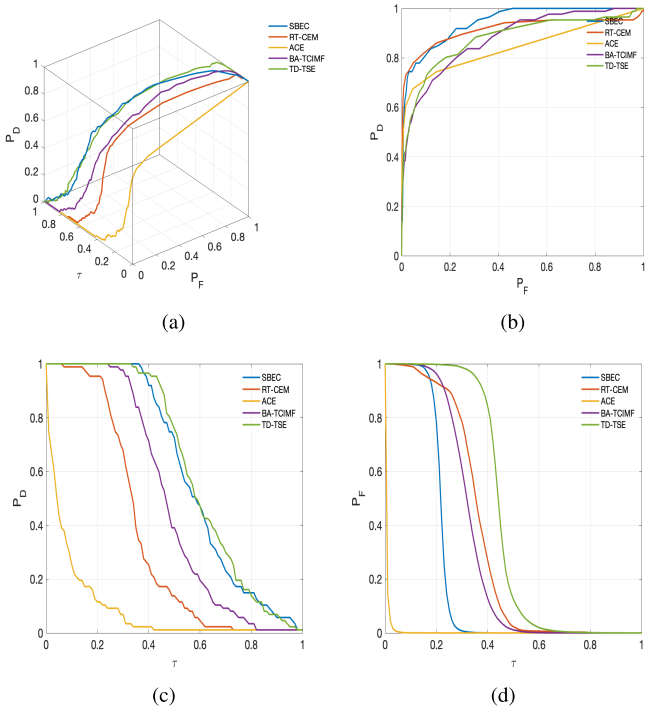

Fig. 12. ROC curves obtained by different methods for Data 3. (a) Three-dimensional ROC curves. (b) ROC curves of (PD, PF). (c) ROC curves of (PD, τ). (d) ROC curves of (PF, τ).

Cores, and 32-GB RAM. The cloud server is composed of three hosts equipped with an 18 core i9-10980K CPU, 128-GB RAM, and 1T disk storage.

Table V shows the comparison of the computational latency between SBEC and CE-SBEC algorithms. If the SBEC algorithm is deployed at the cloud, the data captured at the edge must be uploaded to the cloud before detection. This process does not guarantee real-time performance. Therefore, in the comparative experiments, all task nodes are deployed at the edge when performing the SBEC algorithm. The detection module is deployed at the edge, and the remaining computing nodes are deployed in the cloud when performing the CE-SBEC algorithm. During the target detection process, the CE-SBEC algorithm performs a total of 20 cloud-edge interactions. Table VI presents a comparison of accuracy between the SBEC algorithm and CE-SBEC algorithm when performing target detection on three different datasets. We can observe that there is an insignificant difference in detection accuracy between the SBEC algorithm and the CE-SBEC algorithm. Therefore, the CE-SBEC algorithm significantly reduces target detection computational latency without deteriorating detection accuracy.

TABLE VIII
COMPUTATIONAL LATENCY OF ALL ALGORITHMS

	Algoritms	**Data1**	**Data2**	**Data3**
real-time	CE-SBEC	0.3s	0.28s	0.34s
	RT-CEM	0.27s	0.25s	0.3s
non-real-time	ACE	36s	31s	40s
	BA-TCIMF	116s	108s	128s
	TD-TSE	212s	223s	208s

TABLE IX
LATENCY CORRESPONDING TO DIFFERENT DATA SIZES

Data Size	Latency	Latency Ratio
1GB	114.8s	1
2GB	235.6s	2.05
4GB	473.1s	4.12
8GB	935.6s	8.15

Table VII shows the computational latency of the CE-SBEC algorithm on the three hyperspectral datasets. The fast endmember extraction algorithm proposed in this article significantly reduces the computation time at the cloud. The total latency of transmitting data from the edge to the cloud plus the latency of high-precision detection at the cloud is lower than the latency of real-time detection plus the latency of data update at the edge. Therefore, the total latency of the CE-SBEC algorithm T_J is the computational latency at the edge plus the latency for the edge to receive the results from the cloud and perform the update algorithm, i.e., $T_J = t_J^{\text{detect,edge}} + t_J^{\text{update,edge}}$. According to the results in Table VII, we know that $t_J^{\text{update,edge}} \ll t_J^{\text{detect,edge}}$, indicating that data updates at the edge do not affect the real-time performance of the whole algorithm.

Similarly, we deployed the other algorithms in Section V-C at the edge to conduct runtime testing. Table VIII presents a comparison in terms of latency between the CE-SBEC algorithm and the other algorithms. It is evident that only the CE-SBEC algorithm and the RT-CEM algorithm achieved real-time performance. The CE-SBEC algorithm introduces a minimal increase in latency compared with the RT-CEM algorithm, which does not significantly impact the overall real-time performance. The other algorithms, due to inherent limitations in their computational process, require a complete recalculation of all previously acquired data with each new data capture, resulting in excessively long computational latency.

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To examine the scalability of the CE-SBEC algorithm, we mosaicked real Data 1 to generate large-scale datasets of size 1, 2, 4, and 8 GB, respectively. Experiments were conducted under the condition that the CE-SBEC algorithm performs one cloud-edge interaction for every 500 captured pixels at the edge. The results are presented in Table IX. According to (21), the computational latency of the fast edge update algorithm is determined by $R(m)^{-1}$ and r_{complex}. The total computational latency does not change with the increase of captured data. In other words, regardless of the amount of data captured at the edge, the update time is constant for each cloud-edge interaction. Therefore, this algorithm is scalable for larger data.

VI. Conclusion

In this article, we present a new hyperspectral target detection algorithm named CE-SBEC, which is designed based on cloud-edge collaboration. At the edge, the CE-SBEC algorithm performs the real-time target detection on captured pixels and periodically identifies suspicious pixels, which are then uploaded to the cloud. At the cloud, the algorithm further analyzes the uploaded pixels, especially those complex background pixels, where the accurate energy suppression cannot be achieved by the current edge filters. Subsequently, the analysis results are sent back to the edge for data update, improving the accuracy of the real-time target detection at the edge in the next round. This processing flow represents a completely innovative hyperspectral target detection approach. Our experimental results demonstrate that, compared with state-of-the-art real-time detection algorithms, our proposed CE-SBEC achieves higher accuracy while ensuring real-time performance. With the increasing importance of deep learning in hyperspectral target detection, in future research, we aim to incorporate edge intelligence techniques into deep learning-based hyperspectral target detection to enhance real-time detection efficiency and accuracy at the edge.

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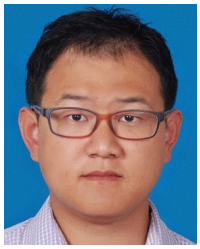

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